

4 (a) electric field strength is the force per unit positive charge (acting on a stationary charge) B1 [1]

(b) (i) $E = V / d$ C1
 $= 1200 / 14 \times 10^{-3}$
 $= 8.57 \times 10^4 \text{ V m}^{-1}$ A1 [2]

(ii) $W = QV$ or $W = F \times d$ and therefore $W = E \times Q \times d$ C1
 $= 3.2 \times 10^{-19} \times 1200$
 $= 3.84 \times 10^{-16} \text{ J}$ A1 [2]

(iii) $\Delta U = mgh$ C1
 $= 6.6 \times 10^{-27} \times 9.8 \times 14 \times 10^{-3}$
 $= 9.06 \times 10^{-28} \text{ J}$ A1 [2]

(iv) $\Delta K = 3.84 \times 10^{-16} - \Delta U$
 $= 3.84 \times 10^{-16} \text{ J}$ A1 [1]

(v) $K = \frac{1}{2}mv^2$ C1
 $v = [(2 \times 3.8 \times 10^{-16}) / 6.6 \times 10^{-27}]^{1/2}$
 $= 3.4 \times 10^5 \text{ m s}^{-1}$ A1 [2]

5 (a) (i) sum of currents into a junction = sum of currents out of junction B1 [1]